

Real Estate CRM

Technology Stack, Dependencies & System Architecture Specification

Project Name:	Real Estate CRM
GitHub Repository:	https://github.com/patturajan045/real-estate-crm
Live Deployment:	Render Cloud Web Service (WSGI / Gunicorn)
Primary Language:	Python 3.10+ (Flask Framework)
Database Engine:	MongoDB 6.0+ (MongoEngine ODM & PyMongo)
Release Specification:	Production Dependencies (requirements.txt)

System Architecture Topology

Real Estate CRM is architected with a decoupled, high-throughput 3-tier topology engineered for scalability, low-latency client interactions, and zero-downtime cloud deployments:

- Presentation Tier (Client Layer):** Pure HTML5, CSS3, and modular ES6+ JavaScript utilizing Bootstrap 5, jQuery DataTables, SweetAlert2, and Chart.js. Requires zero Node.js build pipeline, enabling instant cloud serving.
- Application Tier (Python Flask WSGI):** Centralized Flask application factory registering 10 domain-driven blueprints (Auth, Users, Leads, Projects, Buildings, Units, Bookings, Dashboard, CMS, Notifications). JWT authentication provides stateless session management.
- Persistence Tier (MongoDB Engine):** High-performance document store utilizing MongoEngine ODM for schema constraints and cascading delete safety rules, paired with native PyMongo for fast atomic updates.

1. Python Backend Dependencies (requirements.txt)

The following table provides an exhaustive breakdown of every package specified in requirements.txt, its exact pinned version, architectural purpose, and engineering justification:

Package Name	Version	Architectural Role	Engineering Justification & Usage
Flask	3.0.2	Core Web Framework	Lightweight WSGI framework providing HTTP routing, blueprint modularity, request lifecycle handling, and JSON responses.
mongoengine	0.27.0	Document-Object Mapper	Provides strict typed schema models, UUID primary keys, field validation, unique/compound indexing, and CASCADE/NULLIFY delete rules.
pymongo	4.6.2	Low-Level MongoDB Driver	Official Python driver for MongoDB. Handles socket connection pooling, BSON wire serialization, and fast atomic collection updates.
Flask-JWT-Extended	4.6.0	Security & JWT Auth	Generates and verifies cryptographic HMAC-SHA256 JWT tokens, protecting REST API endpoints via @jwt_required decorators.
Werkzeug	3.0.1	Security & WSGI Tools	Underlying WSGI utility library providing secure enterprise password hashing (pbkdf2:sha256/scrypt) and HTTP request parsing.
python-dotenv	1.0.1	Configuration Loader	Automatically loads environment variables from .env files into os.environ for database URIs and secret keys.
Flask-CORS	4.0.0	Cross-Origin Security	Handles Cross-Origin Resource Sharing (CORS) headers, enabling secure external client and decoupled API integration.
reportlab	>=4.0.0	PDF Document Generator	Programmatic PDF generation engine used to compile publication-ready technical specifications, user manuals, and reports.
gunicorn	>=21.2.0	Production WSGI Server	Production-grade UNIX HTTP server used on Render cloud to serve concurrent worker processes with automatic process recycling.
waitress	>=3.0.0	Windows WSGI Server	Production pure-Python multi-threaded WSGI server for enterprise on-premise Windows deployments.

2. Database Engine: MongoDB & Document Store

Why MongoDB for Real Estate CRM?

- **Natural Hierarchical Modeling:** Real estate inventory naturally follows a nested hierarchy (*Project* → *Building* → *Unit*). Customer leads also embed chronological activity notes (LeadNote) within the lead document, eliminating expensive relational multi-table JOINS.
- **Atomic Check-and-Set Concurrency:** MongoDB's atomic document update operator (update_one(set__status=...)) allows atomic reservation locks in a single round-trip, preventing race-condition double bookings across distributed sales staff.
- **High Throughput & Cloud Portability:** Runs seamlessly locally on localhost:27017 or globally distributed via **MongoDB Atlas** with zero code modifications.

3. Frontend Technology Stack (No-Build Architecture)

The frontend architecture avoids slow, complex JavaScript bundlers (Node.js, Webpack, Vite) in favor of a **zero-build, modular ES6+ architecture** served directly by Flask. This ensures rapid deployment, instantaneous caching, and zero deployment build failures:

Technology / Library	Version	Source / CDN	Role & Implementation
Bootstrap	5.3.3	jsDelivr CDN	Responsive mobile-first grid, modal dialogs, flexbox utilities, and navigation bars.
jQuery	3.7.1	jQuery Official CDN	High-speed DOM traversal, event management, and DataTables binding engine.
jQuery DataTables	1.13.7	DataTables CDN	Interactive tabular display with instant search, multi-column sorting, and custom pagination.
SweetAlert2	11.10.5	jsDelivr CDN	Accessible, beautiful dialog popups, confirmation prompts, and non-blocking toast notifications.
Chart.js	4.4.1	jsDelivr CDN	HTML5 Canvas data visualizations: Leads by Pipeline Stage (Bar) and Inventory Distribution (Donut).
FontAwesome	6.5.1	cdnjs CDN	Vector iconography for buttons, navigation items, status pills, and form inputs.
Google Fonts (Inter)	Latest	Google Fonts	High-legibility geometric sans-serif typeface designed for enterprise dashboard user interfaces.
Modular CRM Core JS	2.0.0	Internal (static/js/)	Decoupled scripts (crm-app.js, auth.js, dashboard.js, leads.js, properties.js, bookings.js, users.js, settings.js).

4. Cloud Deployment Architecture on Render

Render Cloud Web Service Configuration:

- **Service Type:** Python Web Service.
- **Build Command:** pip install -r requirements.txt.
- **Start Command:** gunicorn --bind 0.0.0.0:\$PORT --workers 4 run:app.
- **Entry Point:** [run.py](file:///d:/real_estate_crm/run.py) exports the top-level WSGI callable app = create_app().
- **Environment Configuration:** Managed securely through Render's Environment Variables dashboard:
 - FLASK_ENV=production
 - SECRET_KEY=<secure-random-32-byte-hex>
 - JWT_SECRET_KEY=<secure-random-32-byte-hex>
 - MONGO_URI=mongodb+srv://<user>:<pass>@cluster.mongodb.net/real_estate_crm?retryWrites=true&w=majority
- **Continuous Integration & Deployment:** Connected directly to GitHub repository patturajan045/real-estate-crm for automated builds upon git push to main.

5. Summary of Architectural Benefits

- **Zero-Cost Build Overhead:** Eliminating complex node builds ensures 100% reliable deployment on cloud tiers.
- **Stateless Scalability:** JWT authentication enables seamless horizontal scaling across multiple Gunicorn workers.
- **Data Consistency Guarantee:** Atomic MongoDB operations ensure complete protection against race-condition inventory errors.
- **Audit-Ready Compliance:** Fully documented codebase and comprehensive automated test suite (test_crm.py).